

Vinicius Tikara Venturi Date

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UPDATED: 2026-05

Research Interests

String algorithms; combinatorics on words; de Bruijn sequences; Burrows–Wheeler Transform; repetitiveness measures; compressed indexes.

Education

M.Sc. in Informatics

AUG 2024–AUG 2026 (EXPECTED)

Universidade Federal do Paraná (UFPR) — Curitiba, Brazil

- Dissertation (in progress): *Repetitiveness measures in compressed string indexes: suffixient sets, BWT runs, and de Bruijn sequences* (Advisor: Leandro Miranda Zatesko).

B.Sc. in Computer Science

FEB 2019–DEC 2023

Universidade Federal do Paraná (UFPR) — Curitiba, Brazil

- GPA: 0.917/1.000 (UFPR scale).
- Undergraduate thesis: *The NP-completeness of the clique graph recognition problem* (Advisor: André Luiz Pires Guedes).

Publications and Manuscripts

Peer-Reviewed Publications

Date, V. T. V.; Zatesko, L. M. *On the near-tightness of $\chi \leq 2r$: a general σ -ary construction and a binary case via LFSRs*. Accepted at LATIN 2026 (Springer LNCS, to appear). [arXiv:2512.20598](https://arxiv.org/abs/2512.20598)

Ehrenfried, H. V.; Date, V. T. V.; Todt, E. Exploring graph representation strategies for text classification. *Connection Science*, 2023. [doi:10.1080/09540091.2023.2289832](https://doi.org/10.1080/09540091.2023.2289832)

Preprints and Manuscripts

Date, V. T. V.; Zatesko, L. M. *Successor right-special strings with few Burrows–Wheeler Transform runs*. Manuscript, 2026. [arXiv:2602.20949](https://arxiv.org/abs/2602.20949), [earlier version](#)

Scholarships and Awards

Fundação Araucária MSc Studentship

2025–PRESENT

State Research Foundation of Paraná — Curitiba, Brazil

- Competitive studentship (Edital FA 14/2024) for the UFPR Informatics graduate programme; ranked 1st among M.Sc. proposals from UFPR’s Department of Informatics.

Prêmio Estudante Destaque SBC (SBC Outstanding Student Award)

2023

Sociedade Brasileira de Computação (SBC) — Brazil

- Award from the Brazilian Computer Society recognizing the top graduating student of the UFPR Computer Science cohort.

Maratona SBC de Programação – Brazilian ICPC Regionals

2021, 2023

Sociedade Brasileira de Computação (SBC) — Brazil

- Twice qualified and competed in the Brazilian regional phase of the ICPC-style SBC Programming Marathon.

Research Experience

M.Sc. Student Researcher

AUG 2024–PRESENT

Universidade Federal do Paraná (UFPR) — Curitiba, Brazil

- Main result: near-tightness constructions for $\chi \leq 2r$ (suffixient sets vs. BWT runs), including an LFSR-based binary construction.
- Current focus: de Bruijn-type constructions, successor right-special strings, and low-run BWT structure.
- Built C++ tooling to compute BWT runs and experiment with LFSR-based generators to test conjectures and spot recurring structure.

Undergraduate Researcher

2021–2023

Universidade Federal do Paraná (UFPR) — Curitiba, Brazil

- Contributed to projects on graph-based text representations and educational data integration, including manuscript editing and supporting code; work led to one journal and one conference publication.

Research / Software Engineering Intern (20 h/week)

APR 2021–DEC 2023

Centro de Computação Científica e Software Livre (C3SL), UFPR — Curitiba, Brazil

- Developed a data integration and indicator visualisation platform for the Brazilian National Textbook Program (PNLD/FNDE) using Python, PostgreSQL, Node.js/TypeScript, and dockerized services.
- Implemented REST APIs, an e-mail-based password recovery system, and educational indicators; ran database benchmarks inspired by TPC-H.

Undergraduate Researcher (Scientific Initiation)

2019–2020

Universidade Federal do Paraná (UFPR) — Curitiba, Brazil

- Created and classified programming exercises in the learning tool FARMA-ALG.

Teaching and Mentoring

ICPC Team Coach

2024

Universidade Federal do Paraná (UFPR) — Curitiba, Brazil

- Coached a student team that qualified for the Brazilian ICPC national finals.

Teaching Assistant

2021–2025

Universidade Federal do Paraná (UFPR) — Curitiba, Brazil

- Algorithms / Graph Theory (Mar 2025–Jul 2025): guest lectures on max flow / min cut (Ford–Fulkerson, Edmonds–Karp) and a programming assignment on flow problems.
- ICPC Algorithms (Mar 2023–Dec 2023): co-lectured an ICPC-style problem-solving course; guided students on algorithmic design and implementation.
- Introduction to Programming for Engineers (May 2021–Aug 2021): supported labs and helped design/maintain programming exercises.

Academic Service

Member of B.Sc. thesis examining committee (TCC)

2025

Universidade Federal do Paraná (UFPR) — Curitiba, Brazil

- Examining committee member for a B.Sc. final project (Trabalho de Conclusão de Curso, TCC) in Computer Science.

Technical Skills

- **Programming:** C, C++, Rust, Python.
- **Tools:** Git, Linux, $\text{\LaTeX}$.
- **Languages:** Portuguese (native); English (IELTS Academic 7.5, CEFR C1); Japanese (basic).